Verification guide

Rank-six categorification through multiplicity eight

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1 Scope and authoritative records

The package classifies categorifiability of the 16612 rank-six fusion rings of multiplicity at most eight in the published *Fusion Ring Census*. The current ledger has 16553 general exclusions, 56 unitary-positive rings, one spherical-positive ring admitting no pseudo-unitary categorification, and 2 unresolved rings. Positive based rings, rather than the number of constructions, are counted: the 57 positive rings have 64 constructor records.

The authoritative files are `data/census.json`, `data/decisions.json` and `data/classification_counts.json`. The CSV ledger and paper tables are derived from them. The model catalogue records the source, categorical structure and complete basis permutation for each model. The file `data/unresolved_tensors.json` contains the two complete remaining tensors. General, pivotal, spherical, pseudo-unitary and unitary existence are separate fields.

The census is pinned to *Fusion Ring Census*, revision

`b86eb6dc6eeeafe5fc5f6906d7b074e0322e9726`.

Its rank-six exact-multiplicity counts are $39, 154, 384, 872, 1582, 2768, 3814, 6999$. The compressed tensor file `data/rank6_tables.txt.gz` has SHA-256

`0138c6228dbb06b0861ad38b8dfe67fcd1fc829add80d441c34c09a1afb78675`.

The source and both compressed and uncompressed hashes are recorded in `provenance/repository_source.json`. Import checks fix the input correspondence; completeness of the counting input is supplied by the cited census. The classification does not rerun census enumeration.

2 Checking the release

Use the Python dependencies in `requirements.txt`; C++17 is needed for finite-search helpers and `pdflatex` for rebuilding the documents. From the package root run:

```
python -m pip install -r requirements.txt
python check_manifest.py
python verify.py --jobs 2 --out verification/current/recheck.jsonl
python -O verify.py --jobs 2 --out verification/current/recheck_O.jsonl
python software/current_regressions.py
python -O software/conformal_models.py --verify
python software/release_consistency.py
```

Check the manifest before writing new reports. A new report changes file bytes. The normal and optimized Python checks ensure that correctness does not depend on removable assertions. A subprocess failure has a nonzero return code and must not be reported as a passed check.

The top-level verifier checks the full accepted negative catalogue and positive constructors. The optional `--proofs-only` mode omits positive reconstruction and is therefore not a complete release check. For an independent replay of the accepted continuation certificates use

```
python -O software/continuation_controls.py --jobs 2 \
  --out verification/current/continuation_replay.json
```

This incremental replay does not replace the full-ledger verification. The accepted certificate files retain their frozen tensors, source bindings and input manifests. Their historical filenames are evidence identifiers, not additional current classifications.

A negative checker reconstructs the necessary number fields, finite branch domains, signed quotients, separating vectors and exact inequalities from the full tensor. Numerical optimizers may propose certificates during discovery; their entry points are disabled during negative-certificate checking. An incomplete search, timeout or numerical infeasibility flag cannot establish an exclusion. A successful necessary system cannot establish existence.

These are exact computer-assisted checks, not a proof-assistant formalization. The cited categorical theorems and the proofs of their application remain mathematical inputs. Checksums establish integrity, not the mathematical conclusion.

The multiplicity-five completion. The exclusion of $R^{(6)}_{5,1204}$ first reconstructs its complete 171-profile domain. A forced central product removes 72 profiles; integral cyclic-rotation multiplicities remove the other 99 using 112 certified integer bounds and 99 final exact separating rays. The new proof and solver-disabled replay are under `research/exclusions/R6_5_1204/`:

```
python -O research/exclusions/R6_5_1204/verify_exclusion.py
```

The inherited finite-profile certificate remains part of the proof.

3 The three conformal constructions

The positive registry `data/conformal_positive_models.json` adds $R^{(6)}_{4,22}$, $R^{(6)}_{4,91}$ and $R^{(6)}_{4,629}$. Their models are the neutral components of the module categories for $(B_2)_7 \subset (D_7)_1$, $(C_3)_4 \subset (B_{10})_1$ and $(B_2)_{12} \subset (E_8)_1$, respectively. The source files, exact certificates and independent mathematical audit are organized under `research/structural_models/`.

Each conformal embedding supplies an actual unitary algebra and module category. The parity grading gives a rank-six neutral component. Published action graphs or exact free-induction Gram matrices determine a small set of possible fusion tensors. Local modules, duality, formal-codegree induction and the known center's dimensions or indicators distinguish the required tensor. Every resulting tensor is compared in all 216 entries with the census representative. The proofs do not infer a category from dimensions, a modular invariant, or matching modular data.

The canonical command

```
python -O software/conformal_models.py --verify
```

checks the source bindings and exact tensor identifications. The per-model directories `quartic/`, `c3_model/` and `r629/` contain the reconstruction programs and mathematical proofs, including the deeper graph, Weyl-character and indicator checks. These constructions prove existence. They do not classify all associators on the three rings or identify every archived modular-data matrix in a prescribed labelling.

4 The remaining two problems

The canonical index `research/frontier/` has one directory for each current residual. Each index identifies its retained frozen certificates and replay commands; existing source-bound evidence keeps its original internal paths.

For $R^{(6)}_{4,155}$, the generalized near-group fusion rules give a uniform sphericality and Galois-pseudo-unitarity proof by one norm argument. It replaces the former signed-table search for this ring. Fermion fixed-point traces and odd Gauss sums leave one necessary induction profile, of center rank 52, whose twist spectrum is forced and has exponent 24. They replace the former rank-44 integer-bound certificate. The short exact replay is `research/frontier/R6_4_155/verify_center_cuts.py`. Its pointed center also rules out every braided representative of the Morita class.

For $R^{(6)}_{8,1660}$, the Ising subcategory makes its complement a regular bimodule. Its relative center has rank eighteen and rules $B[t]/(t^2 - At - L)$, where B is the Ising-center ring. Formal-codegree divisibility and dimension-field constraints give 58 labelled choices. A central second-indicator obstruction and a Morita-invariant invertible count leave one based ring, up to exchanging the Ising factors. The proof and exact replay are in `research/frontier/R6_8_1660/relative_center/`. The retained full-center certificates force exponent dividing 48. Fermion orbit constraints and a weighted second-indicator congruence reduce their 1764 labelled profiles to 1727. The tensor involution and homogeneous-packet Vafa equations then leave 1670; the exact replays are in `research/frontier/R6_8_1660/fermion_profiles/` and `research/frontier/R6_8_1660/spin_profiles/`.

None of these necessary alternatives is a categorical model. A claimed future exclusion must cover the entire necessary domain; a claimed model must have a categorical construction and complete tensor identification.

5 Rebuilding the presentation

After checking the retained evidence, regenerate the ledger exports, model catalogue, tables and PDFs with

```
python software/release.py
python software/model_catalog.py
python software/build_paper_tables.py
python build_papers.py
```

Run discovery or regeneration in a working copy, because these commands write files. The build publishes PDFs only after successful compilation. All counts must agree with the decision ledger. The supplement retains accepted proofs, needed dependencies and current residual evidence; obsolete searches for the three conformally realized rings are omitted.